STRUCTURAL OPTIMIZATION

TUNNEL PAVILION

TYPE: Structural Analysis

YEAR: 2025

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FACULTY: Marco Pellegrino, Aaron Potterfield, Matea Pinjusic

CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

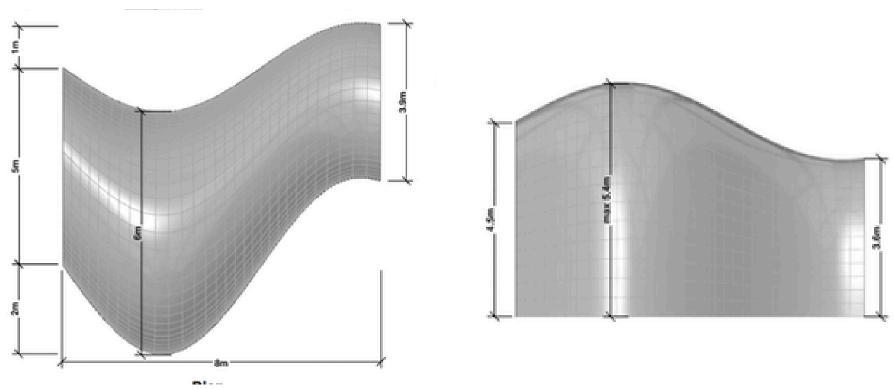
Tunnel Pavilion investigates the design and evaluation of a lattice structure using computational geometry and structural optimization.

The workflow begins with a shell analysis to understand stress distribution and deflection behavior, guiding the subsequent lattice design. The structure is then converted into a lattice system and analyzed under self-weight and wind loads using recycled reinforced glass beam elements.

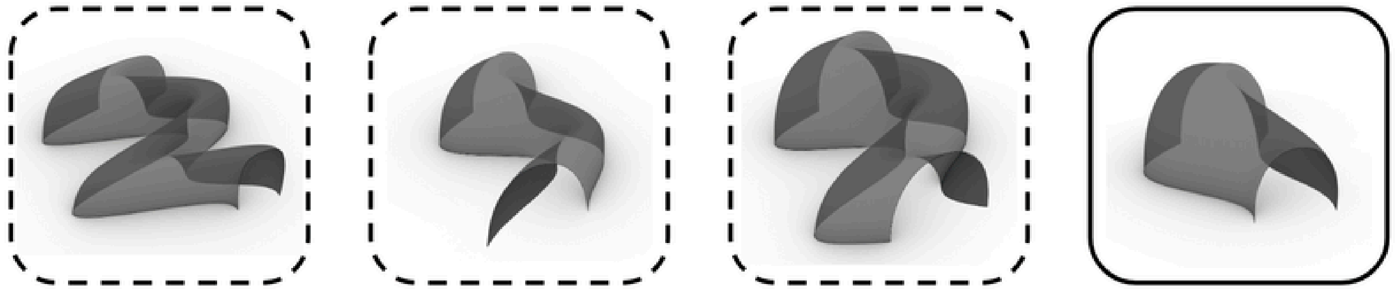
Geometry Exploration → **Shell Analysis** → **Lattice Structure Analysis**

- **Geometry Exploration**: Parametric Surface Equation; Catenary Curves
- **Shell Analysis**: FEM Model; Stress Lines
- **Lattice Structure Analysis**: Lattice Structure Comparison; Optimization Process

GEOMETRY EXPLORATION

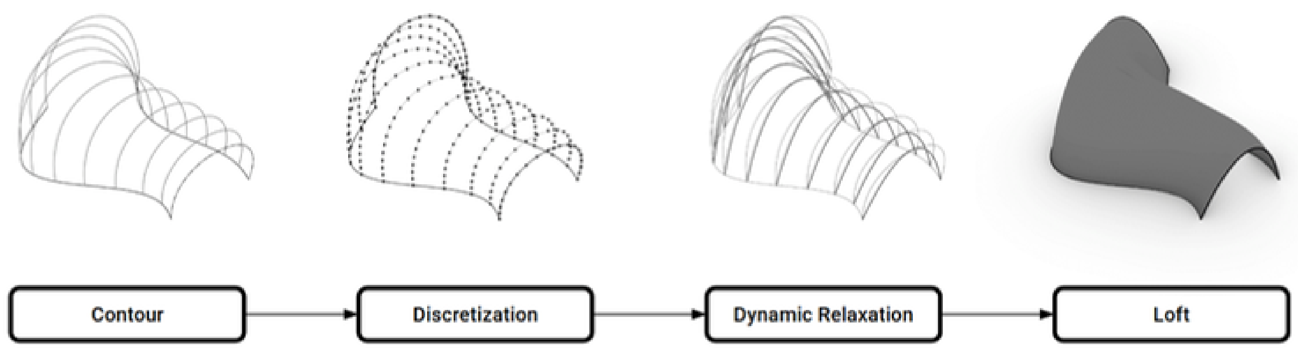


The geometry originates from a parametric surface equation; adjusting parameters like scale factor, pinch strength, and wave amplitude, allowed multiple design variations. The surface geometry was also further reshaped by catenary curves.



scale_factor = 3 pinch_strenght = 0.3 wave_amplitude = 2.8 wave_frequency = 3 u_span = 0.6	scale_factor = 3 pinch_strenght = 0.1 wave_amplitude = 2.5 wave_frequency = 2 u_span = 0.6	scale_factor = 4 pinch_strenght = 0.1 wave_amplitude = 2.5 wave_frequency = 3 u_span = 0.5	**scale_factor = 5** **pinch_strenght = 0.1** **wave_amplitude = 2.25** **wave_frequency = 2** **u_span = 0.4**

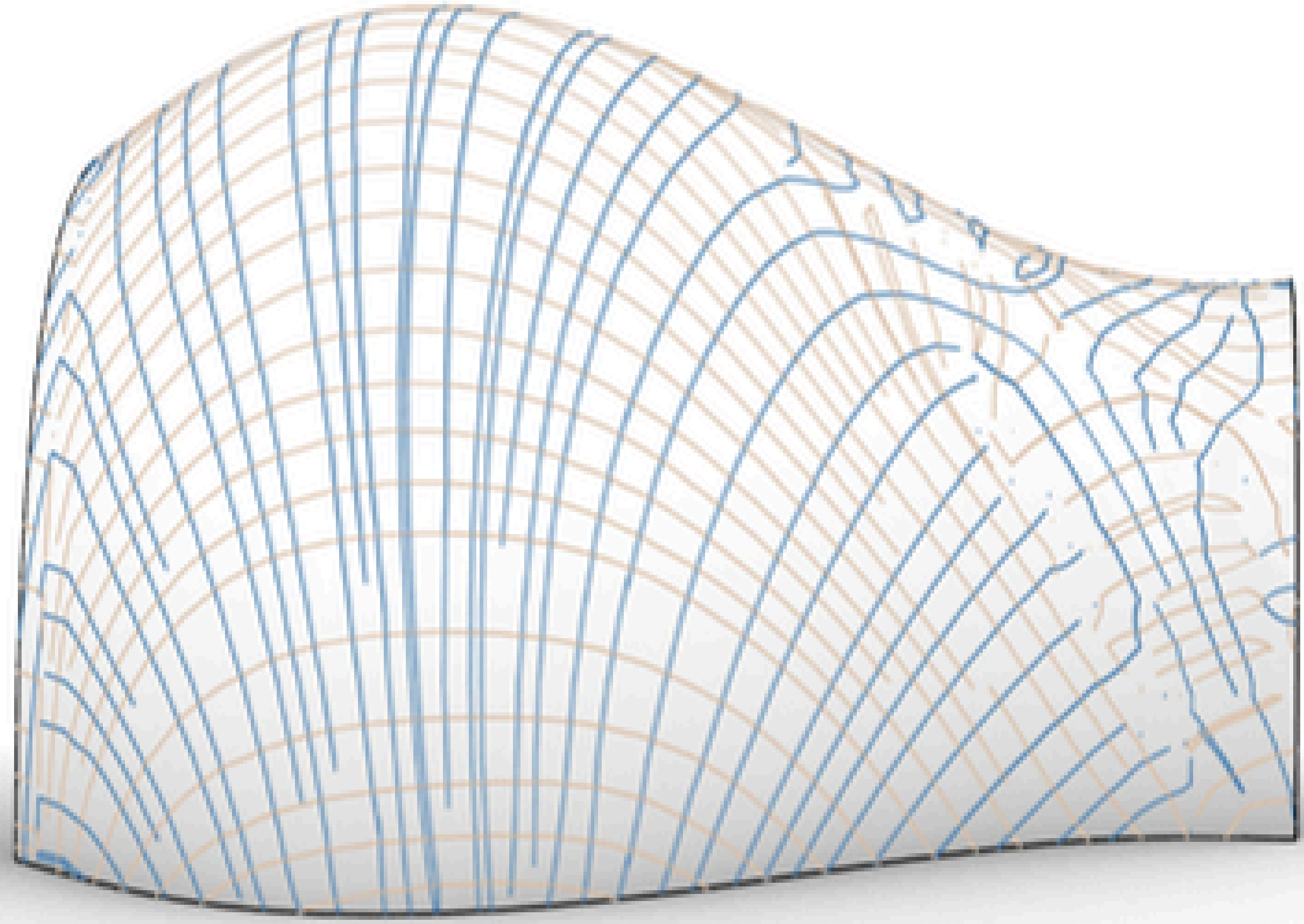
Parametric Surface Equation

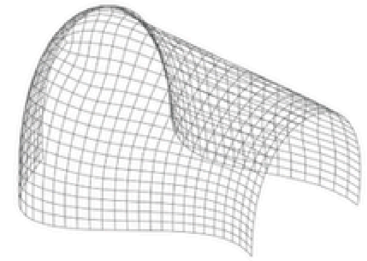


Catenary Curves

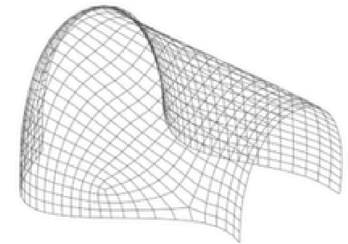
The shell was analyzed in two options: the original mesh and one formed from the catenary curves, reducing von Mises stress at the cost of slightly higher deflection.

Among the tested lattice configurations generated with the Crystallon plug-in, StarTet and BC Cubic lattices demonstrated the best balance between structural performance and fabrication efficiency, requiring the fewest total beam elements, a factor beneficial for 3D printing and fabrication.
Further optimization explored variable lattice thickness and beam elements.



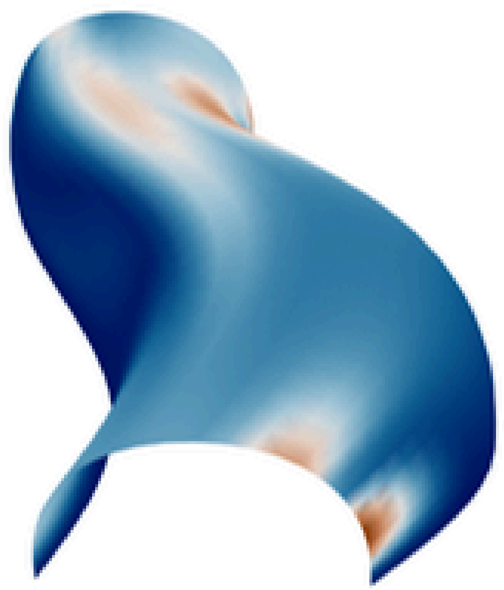


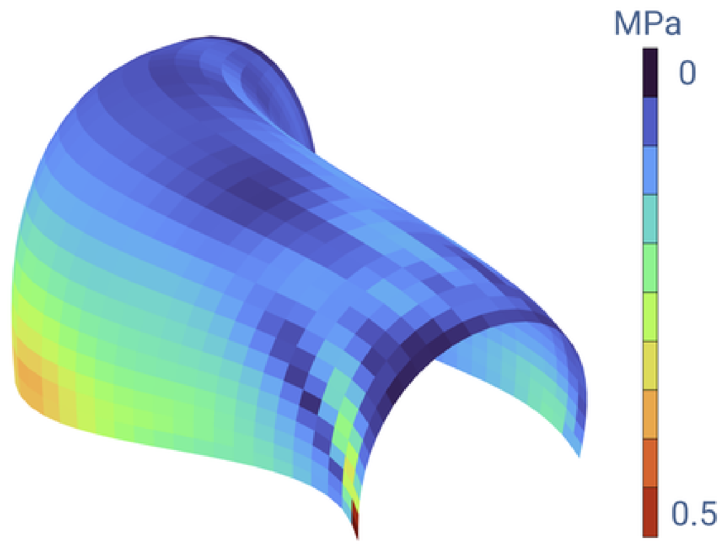
QuadRemesh



QuadRemesh with guided curves

SHELL ANALYSIS





Max Z Deflection

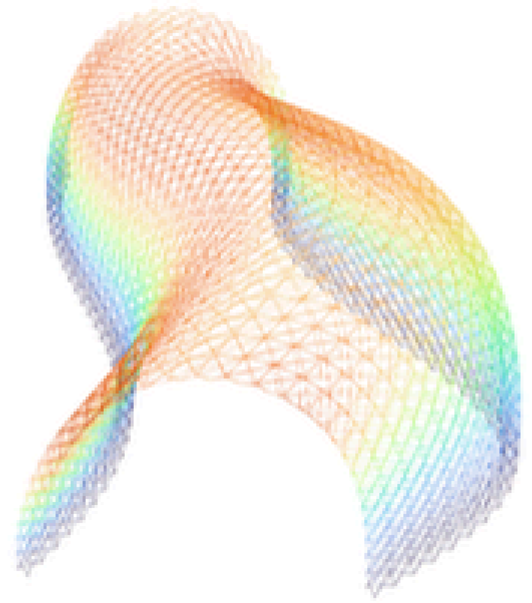
0.66 mm

Von Mises Stress

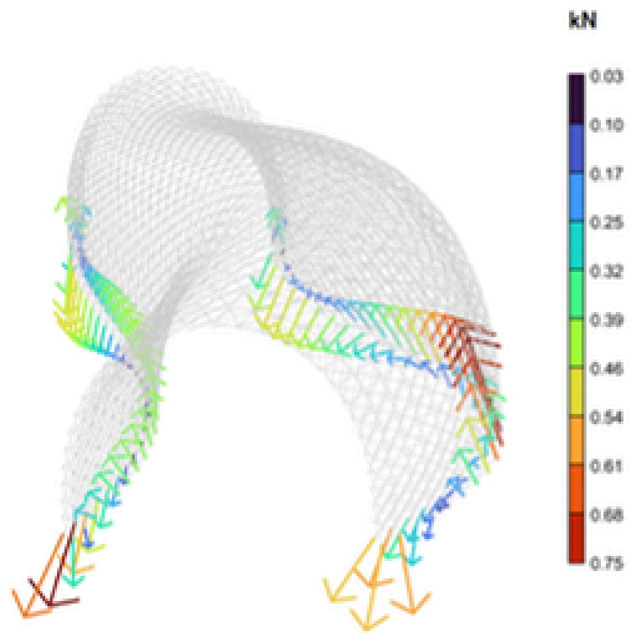
0.68 MPa

LATTICE STRUCTURE ANALYSIS

Deformed Model



Reaction Forces



Tot Element Length	Max Deflection	Von Mises Stress	Utilization Ratio
3262 m	12.5 mm	25.9 MPa	0.29 %